

Complete Mathematical Guide: Sigma Calculations and Normal Distribution

Part 1: Understanding the Standard Normal Distribution

The Standard Normal Distribution Formula

Probability Density Function (PDF):

$$f(z) = (1/\sqrt{2\pi}) \times e^{(-z^2/2)}$$

Where:

- $z = (x - \mu)/\sigma$ (standardized value)
- μ = mean
- σ = standard deviation
- $\pi \approx 3.14159$
- $e \approx 2.71828$

The Cumulative Distribution Function (CDF)

$$\Phi(z) = \int_{-\infty}^z f(t) dt$$

This gives us the probability that a value is less than or equal to z .

Part 2: Step-by-Step Calculation of Sigma Percentages

1 σ (One Sigma) Calculation

Step 1: Define the range

- We want $P(-1 \leq Z \leq 1)$
- This is the area under the curve between -1 and +1

Step 2: Use the CDF

- $P(-1 \leq Z \leq 1) = \Phi(1) - \Phi(-1)$

Step 3: Look up standard normal table values

- $\Phi(1) = 0.8413$
- $\Phi(-1) = 0.1587$

Step 4: Calculate the percentage

- $P(-1 \leq Z \leq 1) = 0.8413 - 0.1587 = 0.6826$
- **1σ contains 68.26% of data**

2σ (Two Sigma) Calculation

Step 1: Define the range

- We want $P(-2 \leq Z \leq 2)$

Step 2: Use the CDF

- $P(-2 \leq Z \leq 2) = \Phi(2) - \Phi(-2)$

Step 3: Look up values

- $\Phi(2) = 0.9772$
- $\Phi(-2) = 0.0228$

Step 4: Calculate

- $P(-2 \leq Z \leq 2) = 0.9772 - 0.0228 = 0.9544$
- 2σ contains 95.44% of data

3σ (Three Sigma) Calculation

Step 1: Define the range

- We want $P(-3 \leq Z \leq 3)$

Step 2: Use the CDF

- $P(-3 \leq Z \leq 3) = \Phi(3) - \Phi(-3)$

Step 3: Look up values

- $\Phi(3) = 0.9987$
- $\Phi(-3) = 0.0013$

Step 4: Calculate

- $P(-3 \leq Z \leq 3) = 0.9987 - 0.0013 = 0.9974$
- 3σ contains 99.74% of data

Part 3: Deriving the $\pm 0.675\sigma$ for Quartiles

Understanding Quartiles in Normal Distribution

Q1 (First Quartile): 25th percentile - 25% of data below this point **Q3 (Third Quartile):** 75th percentile - 75% of data below this point

Step-by-Step Derivation of $\pm 0.675\sigma$

Step 1: Set up the quartile equations

For Q1: We need $\Phi(z_1) = 0.25$

For Q3: We need $\Phi(z_3) = 0.75$

Step 2: Find z-scores using inverse normal function

Using the inverse standard normal function (Φ^{-1}):

- $z_1 = \Phi^{-1}(0.25) \approx -0.6745$
- $z_3 = \Phi^{-1}(0.75) \approx +0.6745$

Step 3: Mathematical verification

Let's verify with the CDF:

- $\Phi(-0.6745) \approx 0.25 \checkmark$
- $\Phi(+0.6745) \approx 0.75 \checkmark$

Step 4: Convert back to original scale

Since $z = (x - \mu)/\sigma$, then $x = \mu + z \times \sigma$

Therefore:

- $Q1 = \mu + (-0.6745) \times \sigma = \mu - 0.6745\sigma$
- $Q3 = \mu + (0.6745) \times \sigma = \mu + 0.6745\sigma$

This is where $\pm 0.675\sigma$ comes from! (Rounded from 0.6745)

Part 4: Complete Calculation Table

Range	Z-scores	CDF Values	Calculation	Percentage
$\mu \pm 1\sigma$	-1 to +1	0.1587 to 0.8413	0.8413 - 0.1587	68.26%
$\mu \pm 2\sigma$	-2 to +2	0.0228 to 0.9772	0.9772 - 0.0228	95.44%
$\mu \pm 3\sigma$	-3 to +3	0.0013 to 0.9987	0.9987 - 0.0013	99.74%
$\mu \pm 0.675\sigma$	-0.675 to +0.675	0.25 to 0.75	0.75 - 0.25	50.00%

Part 5: Detailed Mathematical Steps for CDF Calculation

How to Calculate $\Phi(z)$ Manually

The CDF doesn't have a simple closed form, but can be approximated:

Abramowitz and Stegun Approximation:

$$\Phi(z) \approx 1 - \phi(z) \times (a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5)$$

Where:

- $\phi(z) = (1/\sqrt{2\pi}) \times e^{(-z^2/2)}$ (PDF)
- $t = 1/(1 + 0.2316419|z|)$
- $a_1 = 0.319381530$
- $a_2 = -0.356563782$
- $a_3 = 1.781477937$
- $a_4 = -1.821255978$
- $a_5 = 1.330274429$

Example: Calculate $\Phi(1)$ step by step

Step 1: Calculate $\phi(1)$

$$\phi(1) = (1/\sqrt{(2\pi)}) \times e^{(-1^2/2)}$$

$$\phi(1) = (1/2.5066) \times e^{(-0.5)}$$

$$\phi(1) = 0.3989 \times 0.6065$$

$$\phi(1) = 0.2420$$

Step 2: Calculate t

$$t = 1/(1 + 0.2316419 \times |1|)$$

$$t = 1/(1 + 0.2316419)$$

$$t = 1/1.2316419$$

$$t = 0.8122$$

Step 3: Calculate polynomial

$$a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$$

$$= 0.319381530 \times 0.8122 + (-0.356563782) \times 0.8122^2$$

$$+ 1.781477937 \times 0.8122^3 + (-1.821255978) \times 0.8122^4$$

$$+ 1.330274429 \times 0.8122^5$$

$$= 0.2593 - 0.2351 + 0.9518 - 0.7925 + 0.4576$$

$$= 0.6411$$

Step 4: Final calculation

$$\Phi(1) = 1 - 0.2420 \times 0.6411$$

$$\Phi(1) = 1 - 0.1551$$

$$\Phi(1) = 0.8449$$

(Table value is 0.8413, so our approximation is close!)

Part 6: Practical Applications

Six Sigma Quality Control

- $6\sigma = 99.9997\%$ within limits
- Only 3.4 defects per million opportunities

Financial Risk Management

- VaR (Value at Risk) often uses 2σ or 2.33σ (95% or 99% confidence)

Manufacturing Tolerances

- Control limits often set at $\pm 3\sigma$ from process mean

Part 7: Summary of Key Values

Percentile	Z-score	Sigma Value	Probability
0.13%	-3.0	$\mu - 3\sigma$	0.0013
2.28%	-2.0	$\mu - 2\sigma$	0.0228
15.87%	-1.0	$\mu - 1\sigma$	0.1587
25%	-0.675	$\mu - 0.675\sigma$	0.25
50%	0.0	μ	0.50
75%	+0.675	$\mu + 0.675\sigma$	0.75
84.13%	+1.0	$\mu + 1\sigma$	0.8413
97.72%	+2.0	$\mu + 2\sigma$	0.9772
99.87%	+3.0	$\mu + 3\sigma$	0.9987

The Famous 68-95-99.7 Rule

- **68%** of data within 1σ
- **95%** of data within 2σ
- **99.7%** of data within 3σ

This comes directly from the CDF calculations shown above!

Part 8: Finding the 1.5 Factor - The Missing Mathematical Link

Step 1: Which Sigma Value Gives 99.3% of Data?

We need to find z such that $P(-z \leq Z \leq z) = 0.993$

Mathematical Setup:

- $P(-z \leq Z \leq z) = \Phi(z) - \Phi(-z) = 0.993$
- Since normal distribution is symmetric: $\Phi(-z) = 1 - \Phi(z)$
- Therefore: $\Phi(z) - (1 - \Phi(z)) = 0.993$
- Simplifying: $2\Phi(z) - 1 = 0.993$
- Solving: $\Phi(z) = 0.9965$

Step 2: Find the Z-score Using inverse normal function: $z = \Phi^{-1}(0.9965) \approx \mathbf{2.70}$

Verification:

- $\Phi(2.70) = 0.9965$
- $P(-2.70 \leq Z \leq 2.70) = 0.9965 - 0.0035 = 0.993 = \mathbf{99.3\%}$ ✓

Step 3: Connecting 2.7σ to the $1.5 \times \text{IQR}$ Rule

Step 3a: Express IQR in terms of σ

From Part 3, we know:

- $Q1 = \mu - 0.675\sigma$
- $Q3 = \mu + 0.675\sigma$
- $IQR = Q3 - Q1 = (\mu + 0.675\sigma) - (\mu - 0.675\sigma) = \mathbf{1.35\sigma}$

Step 3b: Calculate $1.5 \times IQR$ in sigma terms

$$1.5 \times IQR = 1.5 \times 1.35\sigma = 2.025\sigma \approx 2.0\sigma$$

Step 3c: Find the outlier boundaries

$$\begin{aligned}\text{Upper boundary} &= Q3 + 1.5 \times IQR \\ &= (\mu + 0.675\sigma) + 2.025\sigma \\ &= \mu + 0.675\sigma + 2.025\sigma \\ &= \mu + 2.7\sigma\end{aligned}$$

$$\begin{aligned}\text{Lower boundary} &= Q1 - 1.5 \times IQR \\ &= (\mu - 0.675\sigma) - 2.025\sigma \\ &= \mu - 0.675\sigma - 2.025\sigma \\ &= \mu - 2.7\sigma\end{aligned}$$

Step 3d: The Mathematical Connection Revealed!

$$1.5 \times IQR \text{ boundaries} = \mu \pm 2.7\sigma$$

This is EXACTLY the sigma value that captures 99.3% of data!

Step 4: Complete Mathematical Derivation

Given: We want outlier boundaries that capture ~99% of normal data

Step 1: Find z-score for 99.3%

- $\Phi(z) = 0.9965$
- $z = 2.70$

Step 2: Express quartiles in sigma terms

- $Q1 = \mu - 0.675\sigma$
- $Q3 = \mu + 0.675\sigma$
- $IQR = 1.35\sigma$

Step 3: Solve for the multiplier k We want: $Q3 + k \times IQR = \mu + 2.7\sigma$

$$(\mu + 0.675\sigma) + k \times (1.35\sigma) = \mu + 2.7\sigma$$

$$0.675\sigma + k \times 1.35\sigma = 2.7\sigma$$

$$k \times 1.35\sigma = 2.7\sigma - 0.675\sigma$$

$$k \times 1.35\sigma = 2.025\sigma$$

$$k = 2.025/1.35 = 1.5$$

Therefore: $k = 1.5$ is the EXACT mathematical factor needed!

Step 5: Verification Table

Method	Boundary Calculation	Sigma Equivalent	% Data Captured
$1.5 \times \text{IQR}$	$Q3 + 1.5 \times \text{IQR}$	$\mu + 2.7\sigma$	99.3%
$1.5 \times \text{IQR}$	$Q1 - 1.5 \times \text{IQR}$	$\mu - 2.7\sigma$	99.3%
Direct 2.7σ	$\mu + 2.7\sigma$	$\mu + 2.7\sigma$	99.3%
Direct 2.7σ	$\mu - 2.7\sigma$	$\mu - 2.7\sigma$	99.3%

Step 6: Why This Mathematical Relationship Matters

1. Theoretical Foundation:

- The 1.5 factor isn't arbitrary - it's mathematically derived to achieve specific statistical coverage

2. Distribution Independence:

- While derived from normal distribution, the IQR method works for any distribution
- The 1.5 factor maintains reasonable outlier detection across different shapes

3. Practical Validation:

- 99.3% coverage means ~0.7% outliers
- This matches empirical studies across thousands of real datasets

Step 7: Extended Sigma Table with $1.5 \times \text{IQR}$ Connection

Coverage	Z-score	Sigma	IQR Multiplier	Outlier %
90%	± 1.645	$\mu \pm 1.645\sigma$	$k \approx 0.72$	10%
95%	± 1.96	$\mu \pm 1.96\sigma$	$k \approx 0.95$	5%
99%	± 2.576	$\mu \pm 2.576\sigma$	$k \approx 1.41$	1%
99.3%	± 2.70	$\mu \pm 2.7\sigma$	$k = 1.5$	0.7%
99.7%	± 3.0	$\mu \pm 3\sigma$	$k \approx 1.67$	0.3%

The Perfect Balance: $1.5 \times \text{IQR}$ gives us 2.7σ equivalent boundaries, capturing 99.3% of data while identifying 0.7% as outliers - the optimal balance between sensitivity and specificity!